Sagar Gharti
Daytona Beach, FL | 479-316-1422 | ghartis@my.erau.edu | *linkedin.com/in/sgharti3/*

Highly motivated aerospace engineer and PhD candidate with over two years of experience in computational modeling of aerodynamics, bio-fluid, and structural analysis problems with computational fluid dynamics (CFD) and finite element analysis (FEA). Apart from the exposure to multidisciplinary projects, I have a strong background in fluid mechanics, thermodynamics, heat transfer, stress analysis, and numerical methods.

Visa Status: F-1 (No sponsorship needed)

SKILLS

Software: STAR-CCM+, SolidWorks, FreeCAD, XFLR5, Microsoft Excel
Languages: Python, MATLAB, Julia
Analysis Methods: CFD, FEA, FSI, LBM

EXPERIENCE

Research Assistant - Embry-Riddle Aeronautical University, FL — **Aug 2024-Present**

- Formulation of an empirical relation to describe convection-diffusion behavior of nutrients in porous flows in interstitial fluid using CFD analysis with STAR-CCM+.
- Simulation of a multiphase (VOF) model of a drop in a vertical wind tunnel with coaxial jets to study the drop dynamics in a varying turbulent field.
- Implementation of RANS and LES turbulence models in the coaxial jet flow field simulation models.
- Implementation of proportional-integrative-derivative (PID) control method in CFD models.
- Fluid-structure analysis to study the bone stresses and their impact on osteoporosis in earth and microgravity.
- Utilizing high-performance computing (HPC) and parallel processing to run the simulations more efficiently.

Teaching Assistant - Embry-Riddle Aeronautical University, FL — **Aug 2024-May 2025**

- Taught undergraduate students to set up an experimental model and collect data in a wind tunnel as part of the experimental aerodynamics lab (compressible and incompressible flows), followed by data analysis and uncertainty analysis.
- Held office hours for the undergraduate senior level incompressible and compressible flow course, along with grading.

Design Engineer – Asha Consulting Group Pvt. Ltd., Nepal — **Feb 2020-Jan 2024**

- Static and dynamic load calculations (thermal, wind, seismic) of heavy machinery structures.
- Stress analysis and structural design of industrial structures and power plants, including pipe racks, silo tanks, cable racks, heavy machinery buildings, etc., using FEA.
- Collaborated with multinational engineering teams like Mitsubishi Heavy Industries (MHI).

Representative Projects:

- Hin-Kong Power Plant, Thailand
- Ghorasal Polash Urea Fertilizer Plant, Bangladesh
- Sakkaide Biomass Power Plant, Japan
- Shintech Industries, US
- Carbon Capture FEED Project, US and Canada

Major responsibilities: Structural analysis, design, load calculation, drawings, quantity estimation, site supervision, quality control, technical presentation, and report writing.

EDUCATION

PhD in Aerospace Engineering (ongoing) — **May 2028 (Expected)**
Embry-Riddle Aeronautical University, Daytona Beach, FL
Master of Science in Aerospace Engineering — **Aug 2024-Dec 2025**
Embry-Riddle Aeronautical University, Daytona Beach, FL
Bachelor of Science in Civil Engineering — **Nov 2014-Oct 2018**
Pulchowk Campus, Tribhuvan University, Nepal

RELEVANT COURSEWORKS

Fluid: Fluid Mechanics, Viscous Flows, Incompressible Aerodynamics, Hydraulics
Thermal: Heat Transfer, Thermodynamics, Boundary Value Problems
Computational: Numerical Methods, Computational Fluid Dynamics, Numerical Linear Algebra, Finite Element Analysis

CLASS PROJECTS

- **Potential Flow Solver Development**: Wrote Python code to build potential flow solvers for thin airfoil theory and source panel methods and compared the results against XFLR5.
- **2-D Plate Heat Transfer Solver Development**: Wrote Python code to solve heat transfer on a plate with various boundary conditions using the finite difference method (FDM).
- **Heat Flux Sensor Building**: Built a heat flux sensor by assembling thermostats on a six-millimeter aluminum cube to measure heat fluxes corresponding to temperatures of up to 350 °C.

REFERENCES

Dr. Michael P. Kinzel, Associate Professor, Embry-Riddle Aeronautical University, FL
Email: kinzelm@erau.edu
Dr. Scott M. Martin, Professor, Embry-Riddle Aeronautical University, FL
Email: martis38@erau.edu
Dr. Ebenezer Gnanamanickam, Associate Professor, Embry-Riddle Aeronautical University, FL
Email: gnanamae@erau.edu
Dr. Andrew Dickerson, Assistant Professor, University of Tennessee, Knoxville, TN
Email: dickerson@utk.edu